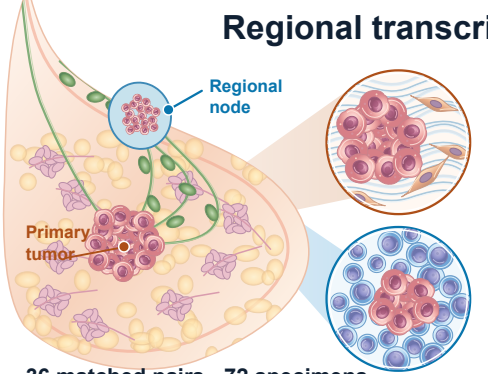
Regional transcriptomic heterogeneity in breast cancer



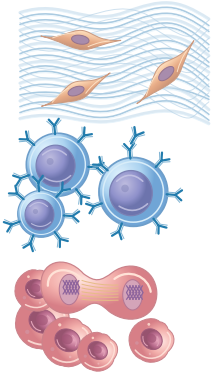
36 matched pairs · 72 specimens

Paired bulk transcriptomics

890 differential genes

562 lower in nodes | 328 higher in nodes

Node versus primary



ECM / stromal programs ↓

B-cell receptor programs ↑

E2F / G2M programs ↑

Molecular readout differences



8/36 pairs

8/36 pairs · 22.2%

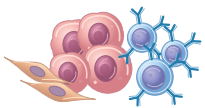
Research PAM50 discordance

(43/50 features)

Luminal A → Luminal B 6 pairs

ROR-S-like scores ↑

Research readouts, not clinical assays



Independent cellular support

5 paired patients

Malignant-cell PLAT and S100A4 ↓



Independent clinical context

METABRIC · 1,890 primary tumors

BCR / E2F / G2M: weak nodal-burden associations

Spearman ρ = 0.098–0.112 · unadjusted

Sampling site matters for transcriptomic interpretation.